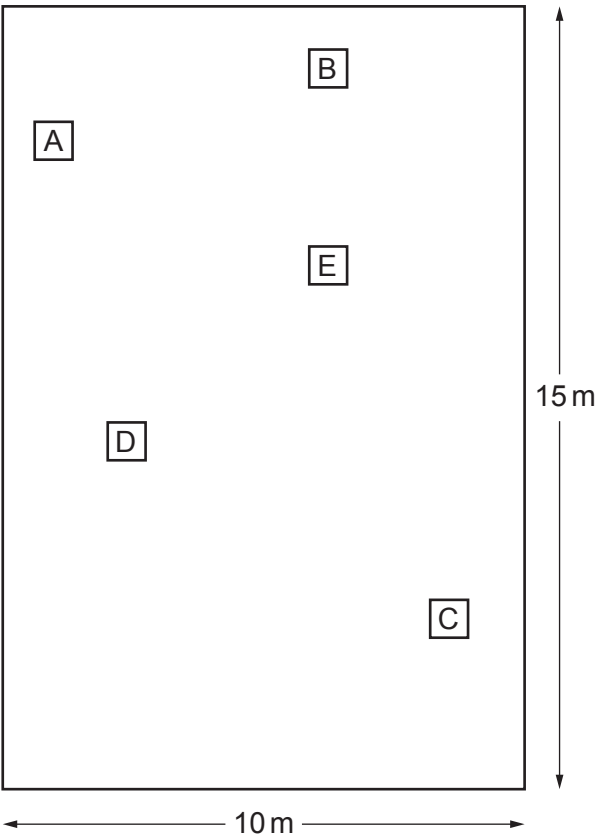


Examiner  
only

6. Some students investigated the number of dandelion plants on a lawn. The diagram shows the lawn and the location of 5 quadrats (**A to E**) which the students had placed at random.



The students counted the number of dandelions in each quadrat and recorded their results in the table below.

| Quadrat | Number of dandelions |
|---------|----------------------|
| A       | 6                    |
| B       | 3                    |
| C       | 4                    |
| D       | 8                    |
| E       | 4                    |

- (a) Calculate the mean number of dandelions per quadrat. [1]

mean number of dandelions = .....

- (b) Calculate the area of the lawn. [1]

area of lawn = ..... m<sup>2</sup>



- (c) Each quadrat has an area of  $1 \text{ m}^2$ . Calculate how many quadrats will fit in the area of the lawn. [1]

number of quadrats = .....

- (d) Use your answers to (a) and (c) to calculate the total number of dandelions on the lawn. [1]

number of dandelions = .....

- (e) The actual number of dandelions on the lawn is 600.  
Use the formula below, and your answer to (d), to calculate the percentage error of your answer above. [2]

$$\text{percentage error} = \frac{\text{number of dandelions} - 600}{600} \times 100$$

percentage error = ..... %

- (f) State **one** improvement to the method used by the students which would give more confidence in their answer for the number of dandelions. [1]

.....

